Evaluation of SpeedyCall Customer Churn Data

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Abstract

Customer churn, the rate at which customers leave a service, is a critical problem in the saturated and highly competitive US telecommunications industry. High churn rates directly impact revenue, and acquiring new customers is significantly more expensive than retaining existing ones. This study addresses this problem by analyzing the "SpeedyCall Customer Churn Data" dataset to identify the key factors driving customer reduction. The primary objectives were to predict churn, evaluate the effect of services on loyalty, analyze the financial impact of churn, and segment customers based on their service usage and account profiles.

The methodology involved a comprehensive data analysis of customer demographic, account, and service data. After cleaning eleven missing values from the dataset, the analysis relied on bivariate visualization techniques. Grouped bar charts were used to compare churn rates across categorical variables such as Contract type, InternetService, TechSupport, and demographic factors like gender, SeniorCitizen, Partner, and Dependents. Box plots were generated to analyze the relationship between churn and key numerical variables, specifically tenure and MonthlyCharges.

The results reveal several significant drivers of churn. Key factors associated with a higher likelihood of churn include having a Month-to-month contract, Fiber optic internet service, and no TechSupport. Demographically, senior citizens and customers without partners or dependents were also more likely to churn. Conversely, Two year contracts and the presence of TechSupport were strongly associated with higher retention and longer customer tenure. Notably, churned customers had a significantly lower median tenure and a higher median MonthlyCharges compared to customers who did not churn.

The conclusion of this study is that customer churn at SpeedyCall is a predictable phenomenon driven by identifiable factors. The findings demonstrate that contract terms, specific service subscriptions(like TechSupport), and internet service type are the most powerful predictors. These insights enable SpeedyCall to move from a reactive to a proactive retention model, allowing for the development of targeted strategies to reduce churn, build customer loyalty, and minimize financial losses.

Content

Introduction

In the highly competitive telecommunications industry, customer churn, that is the rate at which customers leave, is a critical business problem that directly impacts revenue since gaining a new customer is significantly more expensive than retaining an existing one. This telecommunications market is defined by high saturation, rapid technological advancements and increasingly low switching costs for consumers, who can often port their numbers to a new provider with minimal effort. In such an environment, customer loyalty is no longer guaranteed, and even minor dissatisfaction with service, pricing, or support can trigger churn. Therefore, understanding the drivers of churn is a core financial imperative. The analysis of the dataset “SpeedyCall Customer Churn Data" is useful in the analysis of the factors affecting customer churn in a telecommunication company named SpeedyCall in USA.

Objectives:

- Predict customer churn based on their demographic data, service usage and account information.
- Analyze how churn affects revenue and total charges over time.
- To evaluate the effect of specific services on customer loyalty and retention.
- To group customers based on service usage, payment methos and contract period.
- Examine for how long customers typically stay.

The significance of this study is to identify the key demographic and service-related factors of churn and it aims to build a predictive model that can identify at-risk customers before they leave. This allows SpeedyCall to take actions to reduce churn, minimize financial losses, build long-term customer loyalty and increase revenue which will lead it to be successful in the long run.

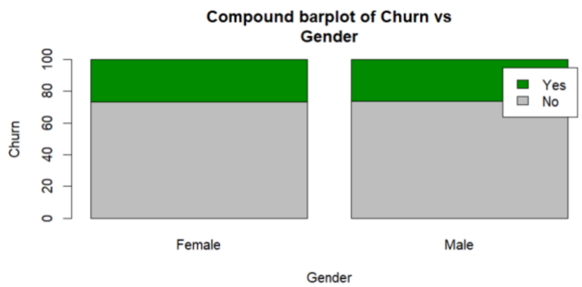
Data

Variable Name	Description
customerID	Customer ID
gender	Whether the customer is a male or a female
SeniorCitizen	Whether the customer is a senior citizen or not (1, 0)
Partner	Whether the customer has a partner or not (Yes, No)
Dependents	Whether the customer has dependents or not (Yes, No)
tenure	Number of months the customer has stayed with the company
PhoneService	Whether the customer has a phone service or not (Yes, No)
InternetService	Customer's internet service provider (DSL, Fiber optic, No)
TechSupport	Whether the customer has tech support or not (Yes, No, No internet service)
Contract	The contract term of the customer (Month-to-month, One year, Two year)
PaymentMethod	The customer's payment method (Electronic check, Mailed check, Bank transfer (automatic), Credit card (automatic))
MonthlyCharges	The amount charged to the customer monthly
TotalCharges	The total amount charged to the customer
Churn	Whether the customer churned or not (Yes or No)

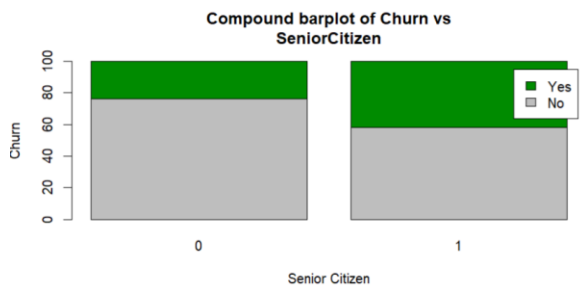
- There were eleven missing values found in the 'Total Charges' variable and the data prepressing methods of cleaning the data were used before the dataset was used for analysis. The R code used is as follows:

```
cols <- c("customerID", "gender", "SeniorCitizen", "Partner", "Dependents",
          "tenure", "PhoneService", "MultipleLines", "InternetService",
          "OnlineSecurity", "OnlineBackup", "DeviceProtection", "TechSupport",
          "StreamingTV", "StreamingMovies", "Contract", "PaperlessBilling",
          "PaymentMethod", "MonthlyCharges", "TotalCharges", "Churn")
cols <- intersect(cols, names(data))
clean_data <- data[complete.cases(data[, cols]), ]
```

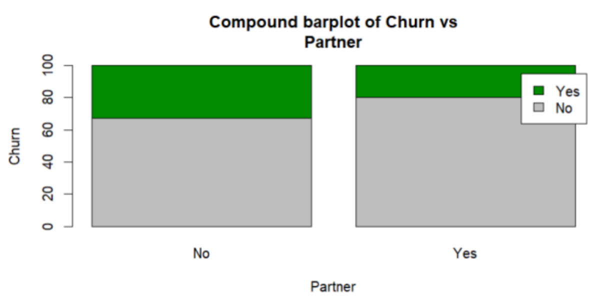
Data Analysis



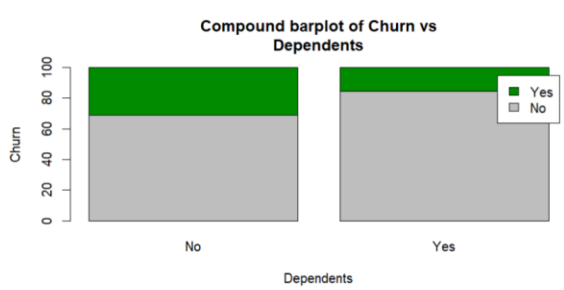
Using this compound bar chart it can be observed that a higher percentage of both males and females have not churned. Further it can be observed that the percentages of both males and females churning is the same.



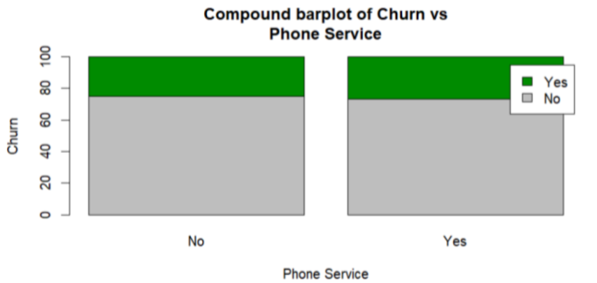
Using this compound bar plot between churn and senior citizen it can be observed that a higher percentage of senior citizens have churned when compared to the people who are not senior citizens. Overall the percentage of people churned is less in both senior citizens and otherwise.



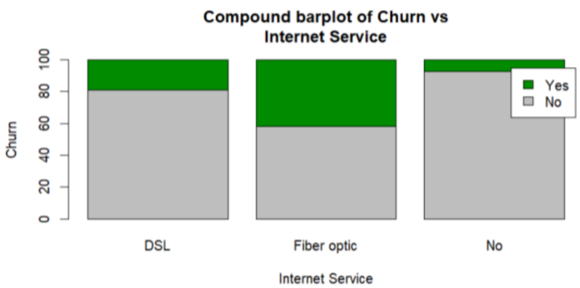
In this compound bar plot it can be observed that the percentage churned is less when the customer has a partner than when the customer doesn't have a partner. The overall percentage churned is less than not churned when considering both categories of partner.



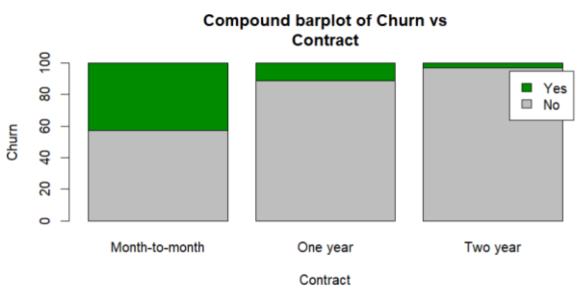
In this compound bar plot the percentage churned is much less in customers who have dependents than customers who do not have dependents. The overall percentage churned is less in both categories of dependents than not churned.



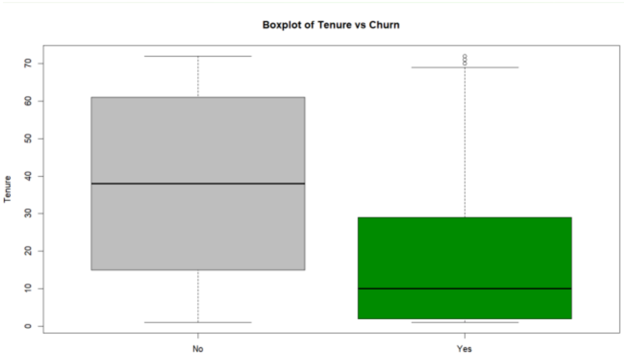
This compound bar chart has almost similar percentages of customers for both types with a phone service and without a phone service. Also, the percentage of cutomers who churned is very much less than who didn't churn.



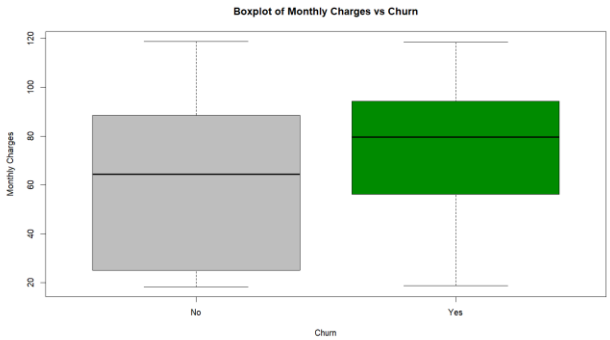
In this compound bar plot it can be observed that the percentage churned is very much less in customers who have DSL as the internet service compared to the customers with Fiber optic internet service.



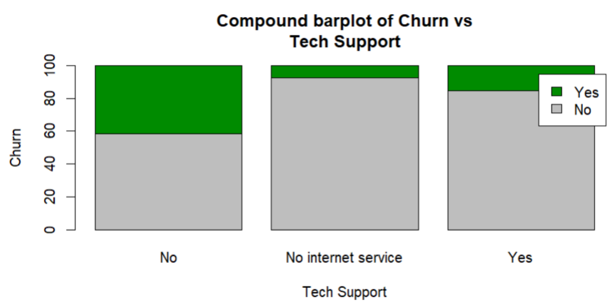
This compound bar plot shows that the least percentage of customers who churned belong to the two year contract category while the highest percentage of customers churned belong to the month-to-month contract category.



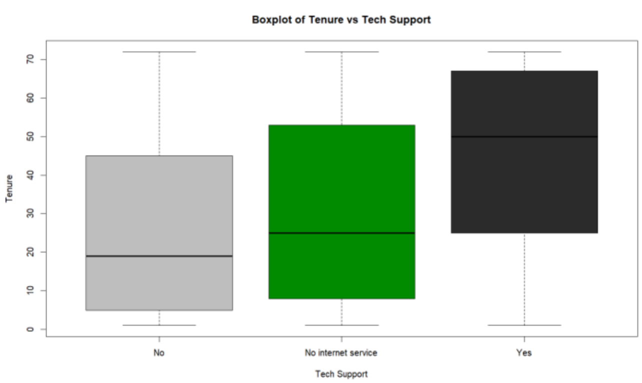
By observing this boxplot it can be studied that the median period of tenure is a lot higher in customers who did not churn compared to the customers who churned. The interquartile range of customers who did not churn is higher than customers who churned. Further, it can also be observed that there are a few outliers in the boxplot where customers churned and they can only be removed if a valid reason can be found by further analysis.



By studying this boxplot it can be observed that the median monthly charge is high in the customers who churned than the customers who didn’t churn. The interquartile range of the customers who didn’t churn is very much higher than the customers who churned.



By observing this compound bar chart it can be observed that the group of customers with no tech support have churned the most while the group with tech support have churned much less than the group without tech support. The least percentage churned is from the group with no internet service.



By observing this boxplot of tenure vs tech support we can observe that the median tenure is the highest in the group of customers who have received tech support while is the lowest in the group with no tech support. It can also be observed that the difference in the medians of the two groups with and without tech support is extremely large. No outliers can be observed in this boxplot.

Variable		Count
Internet Service	DSL	2416
	Fiber optic	3096
	No internet service	1520
Payment Method	Bank transfer	1542
	Credit card	1521
	Electronic check	2365
	Mailed check	1604
Contract	Month-to-month	3875
	One year	1472
	Two years	1685

By using the count tables we can group the customers according to the internet service, payment methos and contract period. The largest group of customers use Fiber optic as the internet service while the least number of customer group has no internet service. It can also be observed that the largest count of customers use electronic check as the payment method while the least number uses credit card as the payment method. When considering contract period, the largest number of customers use month-to-month, while the least number of customers use one year as the contract period.

```
> tenure_freq=ds_freq_table(clean_data,tenure,5)
> tenure_freq
```

Variable: tenure

Bins	Frequency	Cum Frequency	Percent	Cum Percent
1 - 15.2	2459	2459	34.97	34.97
15.2 - 29.4	1099	3558	15.63	50.6
29.4 - 43.6	917	4475	13.04	63.64
43.6 - 57.8	947	5422	13.47	77.1
57.8 - 72	1610	7032	22.9	100
Total	7032	-	100.00	-

This frequency table of tenure can be used to observe for how long customers typically stay. The highest number of customers have tenure for a period of 1 - 15.2 months, while the least number of customers had tenure for a period of 29.4 – 43.6 months.

General Discussion and Conclusion

General Discussion:

- The analysis shows that customer churn is strongly influenced by service-related factors, more than demographic factors such as gender.
- Senior citizens, customers without partners or dependents, and customers with fiber optic internet display higher churn percentages, indicating more dissatisfaction or higher sensitivity to pricing and service issues.
- Month-to-month contract customers churn the most, showing that flexible contracts correlate with lower loyalty, while two-year contract customers are the most stable.
- Customers without tech support churn significantly more, and those who receive tech support stay longer, highlighting tech support as a key retention tool.
- Higher monthly charges are associated with higher churn, suggesting that pricing plays a major role in customers' decision to leave.
- Tenure analysis shows that customers with shorter service durations tend to churn earlier, while long-term customers are more likely to remain.
- Customer groups using electronic checks and fiber optic service represent large segments and should be prioritized for churn-reduction strategies.
- Overall trends emphasize that improving customer experience, lowering perceived costs, and increasing service support can significantly reduce churn.

Conclusion:

- The study identifies clear patterns between churn and factors such as contract type, tech support availability, monthly charges, and internet service type.
- Since churn is highest among month-to-month customers, strengthening long-term contract incentives could effectively improve retention.
- Offering better tech support and customer service would address the groups most at risk of leaving.
- Managing pricing structures, especially for fiber optic customers, can reduce dissatisfaction and prevent revenue loss.
- Predictive insights from this dataset can help SpeedyCall target at-risk customers early, implement personalized retention strategies, and enhance overall customer loyalty.
- By prioritizing service quality, support, and contract-based incentives, SpeedyCall can significantly reduce churn and improve long-term business sustainability.